

AI – ASSISTED FULL STACK WEB DEVELOPMENT

Week 4 — Backend & API Development

1. Overview of the Week

This week you build the engine room of your app. You will set up a backend server using Express.js (Node) or FastAPI (Python), design and implement at least 6 REST API endpoints, test every endpoint in Postman or Thunder Client, and connect your frontend to the backend so real data flows between them. By the end of this week, your app is no longer static — it is a full-stack app with live data.

2. What You Will Do This Week

- **Initialise the backend:**
 - **Node/Express path:** Create /backend folder, `npm init -y`, install `express`, `cors`, `dotenv`, `nodemon`. Create a basic server in `server.js` listening on port 5000.
 - **Python/FastAPI path:** Create /backend, set up a virtual environment (`python -m venv venv`), install `fastapi`, `uvicorn`, `python-dotenv`. Create a basic app in `main.py`.
- **Design 6+ REST endpoints** around your core data model. Example for a task manager app:
 - GET /api/tasks — list all tasks
 - GET /api/tasks/:id — get a single task
 - POST /api/tasks — create a task
 - PUT /api/tasks/:id — update a task
 - DELETE /api/tasks/:id — delete a task
 - GET /api/tasks/search?q=... — search tasks
- **Use proper HTTP methods and status codes:** 200 for successful GET/PUT, 201 for POST create, 204 for DELETE, 400 for validation errors, 404 for not found, 500 for server errors.
- **Add basic error handling middleware** (Express) or exception handlers (FastAPI).
- **Use in-memory data for now** — a simple array or dictionary. Real database comes in Week 5.
- **Test every endpoint in Postman or Thunder Client:** create a collection with 6+ requests covering each endpoint. Export the collection as a JSON file.
- **Connect frontend to backend:** Replace all hardcoded/mock data in your frontend with real API calls using `fetch` or `axios`. Handle loading states (show your Loader component) and error states (show a Toast). Configure CORS on the backend to allow your frontend origin.

3. Learning Outcomes for the Week

Week Outcome	Mapped to CLO
Set up a backend server with Express or FastAPI and clean project structure	CLO-2
Design and implement 6+ REST endpoints with correct HTTP methods and status codes	CLO-2
Test every endpoint using Postman or Thunder Client with documented results	CLO-2
Connect the frontend to the backend with live API calls, loading, and error states	CLO-1, CLO-2

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4. Resources

4.1 YouTube Videos

#	Title / Content	Channel & Link
1	Node.js and Express for Beginners	freeCodeCamp — https://www.youtube.com/watch?v=Oe421EPjeBE
2	Build a REST API with Node.js, Express, and MongoDB	Traversy Media — https://www.youtube.com/watch?v=fgTGADljAeg
3	FastAPI Course for Beginners	freeCodeCamp — https://www.youtube.com/watch?v=tLKKmouUams
4	What Is REST API? Explained in 100 Seconds	Fireship — https://www.youtube.com/watch?v=-MTSQjw5DrM
5	Postman Tutorial — Full Beginner Guide	Traversy Media — https://www.youtube.com/watch?v=VywxIQ2ZXw4
6	Thunder Client — Lightweight API Tester	Ahmad Awais — https://www.youtube.com/watch?v=AbCTlemwZ1k
7	CORS in 100 Seconds	Fireship — https://www.youtube.com/watch?v=4KHiSt0oLJ0
8	Fetch API Tutorial for Beginners	The Net Ninja — https://www.youtube.com/watch?v=cuEtnrL9-H0
9	Express Middleware Explained	Web Dev Simplified — https://www.youtube.com/watch?v=IY6icfhap2o
10	HTTP Status Codes Explained	Web Dev Simplified — https://www.youtube.com/watch?v=5b9nx_w-B2s

4.2 Blogs and Articles

#	Title	Source & Link
1	Express.js Getting Started Guide	Express Official — https://expressjs.com/en/starter/installing.html
2	FastAPI Tutorial — First Steps	FastAPI Docs — https://fastapi.tiangolo.com/tutorial
3	REST API Design Best Practices	MDN Web Docs — https://developer.mozilla.org/en-US/docs/Glossary/REST
4	HTTP Status Codes Reference	MDN — https://developer.mozilla.org/en-US/docs/Web/HTTP/Status
5	Build a RESTful API with Node and Express	DigitalOcean Community — https://www.digitalocean.com/community/tutorials/nodejs-express-routing
6	Postman Learning Center	Postman Docs — https://learning.postman.com
7	CORS Explained Simply	MDN Web Docs — https://developer.mozilla.org/en-US/docs/Web/HTTP/CORS
8	Error Handling in Express	Express Docs — https://expressjs.com/en/guide/error-handling.html
9	Axios vs Fetch — Which to Use	LogRocket Blog — https://blog.logrocket.com/axios-vs-fetch-best-http-requests
10	REST API Naming Conventions	REST API Tutorial — https://restfulapi.net/resource-naming

4.3 MOOCs / Courses

#	Course	Platform & Link
1	Back End Development and APIs Certification	freeCodeCamp (Free) — https://www.freecodecamp.org/learn/back-end-development-and-apis
2	Full Stack Open — Part 3 (Node & Express)	University of Helsinki (Free) — https://fullstackopen.com/en/part3
3	APIs	Coursera (Meta) — https://www.coursera.org/learn/apis
4	FastAPI — The Complete Course	FastAPI Official Learning Resources — https://fastapi.tiangolo.com/learn

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5. Deliverables

Deliverable 1: Working REST API in GitHub Repo

- Backend folder in your GitHub repo with a clean structure.
- Minimum **6 working endpoints** covering CRUD (GET list, GET single, POST, PUT/PATCH, DELETE) plus at least one additional endpoint (search, filter, or similar).
- Each endpoint must return correct HTTP status codes and JSON responses.
- Error handling middleware/exception handlers in place.
- .env file exists but is gitignored; a .env.example file shows required env variable names.
- README updated with a "How to run backend locally" section.

Deliverable 2: Postman/Thunder Client Collection Export

- Export your API testing collection as a JSON file.
- Collection must include one request per endpoint (minimum 6).
- Each request must have a saved example response.
- File name: W4_APICollection_[InternID].json

Deliverable 3: Frontend Connected to Backend — Demo Screenshot PDF

- 2–3 screenshots showing the frontend displaying data from the backend.
- One screenshot of the Network tab in Chrome DevTools showing successful API calls (status 200).
- Add brief captions under each screenshot.
- File name: W4_FrontendBackendConnection_[InternID].pdf

6. Submission Instructions

- Submit W4_APICollection_[InternID].json and W4_FrontendBackendConnection_[InternID].pdf via the **TBI-GEU LMS Assignment box for Week 4**.
- Submit the **GitHub repo URL** (showing both frontend and backend code) in the assignment link field.
- Submit the **Weekly Progress Report** via the TBI-GEU LMS Form by **Sunday 11:59 PM IST**.